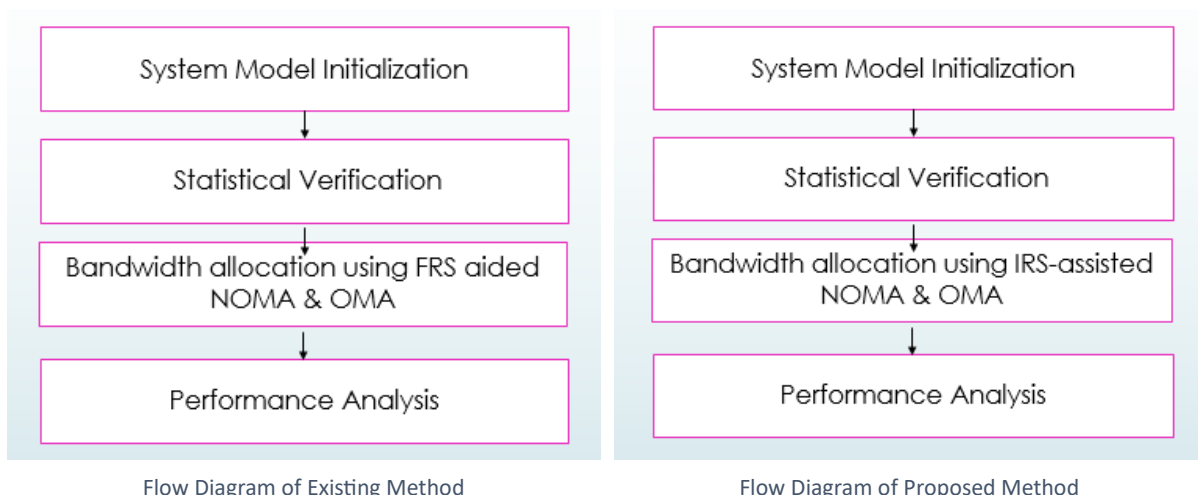


Downlink and Uplink Intelligent Reflecting Surface Aided Networks: NOMA and OMA

Abstract:

Intelligent Reflecting Surfaces (IRS) have emerged as a promising technology to enhance wireless communication by dynamically reconfiguring the propagation environment. This paper investigates the performance of IRS-aided downlink and uplink networks under Non-Orthogonal Multiple Access (NOMA) and Orthogonal Multiple Access (OMA) schemes. The primary objective is to improve coverage, outage probability, ergodic rate, and diversity order by deploying IRS to assist cell-edge users in communication with the base station. New channel statistics are derived for the base station-IRS-user device (BS-IRS-UD) link under Nakagami-m fading, enabling a detailed performance analysis. Closed-form expressions for outage probability and ergodic rate are obtained, followed by asymptotic approximations to determine diversity order and high-SNR slope. Results demonstrate that IRS significantly improves system performance compared to full-duplex decode-and-forward relay (FDR)-aided networks, especially in the high-SNR regime. The study further highlights that diversity order depends on the number of IRS reflecting elements and fading parameters, whereas the high-SNR slope remains independent of these factors. Simulation results validate the theoretical analysis, confirming the advantages of IRS in optimizing NOMA and OMA networks. This work provides valuable insights into IRS deployment strategies for next-generation wireless communication systems.



Conventional Orthogonal Multiple Access (OMA) and Non-Orthogonal Multiple Access (NOMA) are widely used techniques in wireless communication, but both face limitations such as inefficient spectrum utilization, inter-user interference, fairness issues, and poor adaptability in challenging environments. To address these challenges, this work explores the integration of Intelligent Reflecting Surfaces (IRS) with OMA and NOMA in both downlink and uplink scenarios. By strategically deploying IRS to optimize signal reflection, the system enhances spectral efficiency, user fairness, and overall network performance. The proposed approach jointly optimizes IRS phase shifts and power allocation to maximize system sum rate and energy efficiency. In the downlink case, IRS reflection coefficients are adjusted under power constraints to improve achievable sum rate, while in the uplink scenario, user power control is optimized alongside IRS-assisted signal enhancement to minimize interference and improve reception quality.

Equipments:

1. **Hardware:** MATLAB R2022b.
2. **Software:**
 - **Operating Systems:** Windows.
 - **Processors:** Any Intel or AMD x86-64 processor.
 - **Disk:** 2.9 GB of HDD space for MATLAB only and 5-8 GB for a typical installation.
 - **RAM:** 4 GB

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